

Problem 4:

a) A set $\mathcal{M}$ is convex if for any two elements $P_1, P_2 \in \mathcal{M}$ and any $\alpha \in [0, 1]$, the following is satisfied $\alpha P_1 + (1-\alpha)P_2 := P \in \mathcal{M}$, i.e., P is also a probability measure. Let $P_1, P_2 \in \mathcal{M}$ and $\alpha \in [0, 1]$. To prove that $P := \alpha P_1 + (1-\alpha)P_2$ is a probability measure, we need to show:

(i) Well-definedness incl non-negativity ($P(A) \in [0, 1] \quad \forall A \in \mathcal{F}$): For all $A \in \mathcal{F}$, we have $P_1(A) \in [0, 1]$ and $P_2(A) \in [0, 1]$. Since $\alpha \in [0, 1]$, it follows that $1-\alpha \in [0, 1]$. Therefore $P(A) = \alpha P_1(A) + (1-\alpha)P_2(A) \geq 0$ and also $P(A) \leq \alpha(1) + (1-\alpha)(1) = 1$. Thus, P is a well-defined mapping from $\mathcal{F}$ to $[0, 1]$.

(ii) Normalization: $P(\Omega) = \alpha P_1(\Omega) + (1-\alpha)P_2(\Omega) = \alpha(1) + (1-\alpha)(1) = 1$

(iii) σ -additivity: Let $(A_n)_{n \in \mathbb{N}} \in \mathcal{F}$ with $A_i \cap A_j = \emptyset \quad \forall i \neq j$. Then, we have:

$$\begin{aligned} P\left(\bigcup_{n \in \mathbb{N}} A_n\right) &= \alpha P_1\left(\bigcup_{n \in \mathbb{N}} A_n\right) + (1-\alpha)P_2\left(\bigcup_{n \in \mathbb{N}} A_n\right) \stackrel{(P_1, P_2 \in \mathcal{M})}{=} \alpha \sum_{n \in \mathbb{N}} P_1(A_n) + (1-\alpha) \sum_{n \in \mathbb{N}} P_2(A_n) \\ &= \sum_{n \in \mathbb{N}} \alpha P_1(A_n) + \sum_{n \in \mathbb{N}} (1-\alpha)P_2(A_n) = \sum_{n \in \mathbb{N}} (\alpha P_1(A_n) + (1-\alpha)P_2(A_n)) = \sum_{n \in \mathbb{N}} P(A_n). \end{aligned}$$

Thus, $P \in \mathcal{M}$, and $\mathcal{M}$ is convex. $\square$

b) To prove that the set of extreme points of $\mathcal{M}$ is given by the Dirac measures, we need to prove:

(i) Dirac measures are extreme points. We first need to prove that $\delta_w \in \mathcal{M}$

by showing:

(1) (Well-defined): $\forall A \in \mathcal{F}, \delta_w(A) \in \{0, 1\} \subseteq [0, 1]$.

(2) (Normalization): Since $w \in \Omega$, then $\delta_w(\Omega) = 1$.

(3) (σ -additivity): Let $(A_n)_{n \in \mathbb{N}} \in \mathcal{F}$ with $A_i \cap A_j = \emptyset \quad \forall i \neq j$. Since $A_i \cap A_j = \emptyset$,

w can belong to, at most, one set A_k :

• If $\exists k \in \mathbb{N} \mid w \in A_k$, then $w \in \bigcup_{n \in \mathbb{N}} A_n$ and $\delta_w\left(\bigcup_{n \in \mathbb{N}} A_n\right) = 1$. Since A_k is unique $\sum_{n \in \mathbb{N}} \delta_w(A_n) = \delta_w(A_k) + \sum_{i \neq k} \delta_w(A_i) = 1$.

• If $\nexists k \in \mathbb{N} \mid w \in A_k$, then $w \notin \bigcup_{n \in \mathbb{N}} A_n$ and $\delta_w\left(\bigcup_{n \in \mathbb{N}} A_n\right) = 0 = \sum_{n \in \mathbb{N}} \delta_w(A_n)$.

In both cases $\delta_w\left(\bigcup_{n \in \mathbb{N}} A_n\right) = \sum_{n \in \mathbb{N}} \delta_w(A_n)$

Since δ_w is well-defined and satisfies the axioms (2) and (3), δ_w is a probability measure, i.e., $\delta_w \in \mathcal{M}$.

Assume that $\delta_w = \alpha P_1 + (1-\alpha)P_2$ for some $\alpha \in [0, 1]$ and $P_1, P_2 \in \mathcal{M}$, for any $w \in \Omega$. Let $A \in \mathcal{F}$ st. $w \in A$, then $1 = \delta_w(A) = \alpha P_1(A) + (1-\alpha)P_2(A)$ if and only if $P_1(A) = P_2(A) = 1$ since $P_1(A), P_2(A)$ and $\alpha \in [0, 1]$.

Since $P_1, P_2 \in \mathcal{M}$, then $P_1(A^c) = 0$, $P_2(A^c) = 0$. Since $\omega \in A$, then $\omega \notin A^c$ and $\delta_\omega(A^c) = 0$. Thus, for any $A \in \mathcal{F}$,

$$P_1(A) = P_2(A) = \begin{cases} 1, & \omega \in A \\ 0, & \omega \notin A \end{cases}$$

This implies $\delta_\omega = P_1 = P_2$, i.e., δ_ω is an extreme point.

(ii) Only Dirac measures are extreme points. Let $P \in \mathcal{M}$ be a probability measure that is not a Dirac measure. Then, we can prove that $\exists E \subset \Omega \mid P(E) \in (0, 1)$:

By contradiction: Suppose $\nexists E \subset \Omega \mid P(E) \in (0, 1)$, then $\forall E \subset \Omega \mid P(E) \in \{0, 1\}$. Since Ω is countable, $1 = P(\Omega) = \sum_{\omega \in \Omega} P(\{\omega\})$ (σ-additivity). Since $P(\{\omega\}) \in \{0, 1\}$, then exactly one element $\omega^* \in \Omega$ must have $P(\{\omega^*\}) = 1$, and all others 0. This implies $P = \delta_{\omega^*}$, contradicting that P is not a Dirac measure. Therefore, $\exists E \subset \Omega \mid P(E) \in (0, 1)$.

We can construct two new (conditional) probability measures (conditional on E and E^c):

$$P_1(A) = \frac{P(A \cap E)}{P(E)} \quad \text{and} \quad P_2(A) = \frac{P(A \cap E^c)}{P(E^c)} \quad (A \in \mathcal{F})$$

$$\text{Since } P_1(E) = \frac{P(E \cap E)}{P(E)} = \frac{P(E)}{P(E)} = 1 \neq 0 = \frac{P(\emptyset)}{P(E)} = \frac{P(E \cap E^c)}{P(E^c)} = P_2(E),$$

then $P_1 \neq P_2$. Let $\alpha = P(E) \in (0, 1)$, and then:

$$\begin{aligned} P(A) &= P(A \cap E) + P(A \cap E^c) = P(E) \frac{P(A \cap E)}{P(E)} + P(E^c) \frac{P(A \cap E^c)}{P(E^c)} \\ &= \alpha P_1(A) + (1 - \alpha) P_2(A) \end{aligned}$$

Therefore, P can be written as a combination of two distinct probability measures in $\mathcal{M}$. By definition, P cannot be an extreme point.

We prove that every $P \in \mathcal{M}$ can be represented as a "mixture" of Dirac measures, i.e., $P = \sum_{\omega \in \Omega} \alpha_\omega \delta_\omega$ with $\alpha_\omega \geq 0$ and $\sum_{\omega \in \Omega} \alpha_\omega = 1$:

Let $P \in \mathcal{M}$ and let $\alpha_\omega = P(\{\omega\}) \in [0, 1] \quad \forall \omega \in \Omega$. Then, any $A \in \mathcal{F}$ can be represented as $A = \bigcup_{\omega \in A} \{\omega\}$ (as Ω is at most countable) and: by σ-additivity

$$P(A) = P\left(\bigcup_{\omega \in A} \{\omega\}\right) = \sum_{\omega \in A} P(\{\omega\}) = \sum_{\omega \in A} P(\{\omega\})(1) + \sum_{\omega \notin A} P(\{\omega\})(0) = \sum_{\omega \in A} \alpha_\omega(1) + \sum_{\omega \notin A} \alpha_\omega(0) = \sum_{\omega \in \Omega} \alpha_\omega \delta_\omega(A)$$

$$\text{with } P(\Omega) = \sum_{\omega \in \Omega} \alpha_\omega \delta_\omega(\Omega) = \sum_{\omega \in \Omega} \alpha_\omega(1) = \sum_{\omega \in \Omega} \alpha_\omega = \sum_{\omega \in \Omega} P(\{\omega\}) = P(\Omega) = 1$$

Therefore $P = \sum_{\omega \in \Omega} \alpha_\omega \delta_\omega$ with $\alpha_\omega \geq 0$ and $\sum_{\omega \in \Omega} \alpha_\omega = 1$.